SQL – Practical & Analytical (Star + Functions + Joins)

Question 1: Multiple Choice

1. What does CONCAT('Mr. ', name) do?

A) Removes spaces

B) Combines strings ✅

C) Converts to uppercase

D) Extracts text

2. What is the output of SUBSTRING('Database',1,4)?

A) Data ✅

B) Base

C) Datab

D) ase

3. Which function removes spaces?

A) LENGTH

B) TRIM ✅

C) ROUND

D) CONCAT

4. What does EXTRACT(YEAR FROM order_date) return?

A) Full date

B) Month

C) Year ✅

D) Day

5. COUNT(*) is used to:

A) Sum values

B) Count rows ✅

C) Find max

D) Find average

6. DATE_TRUNC returns a full date truncated to a specific unit.

✅ True

2. AVG() works only on text values.

❌ False

8. COALESCE returns the first non-null value.

✅ True

9. FLOOR(4.8) = 5

❌ False

10. NOW() returns date and time.

✅ True

11. This command displays student name with class name.

```
SELECT s.name, c.name
FROM students s
JOIN classes c ON s.class_id = c.id;
```

✅ True

12. This command displays student names with subjects.

```
SELECT s.name, sub.title
FROM students s
JOIN student_subjects ss ON s.id = ss.students_id
JOIN subjects sub ON ss.subject_id = sub.id;
```

✅ True

13. Relationship between students and classes:

✅ One-to-Many

14. Relationship between students and subjects:

✅ Many-to-Many

1. Display company names in uppercase. عرض النصوص بأحرف كبيرة

SELECT **UPPER**(company_name) FROM customers;

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2. Display first 3 characters of contact_name with "Mr." دمج نصين

SELECT **CONCAT**('Mr. ', SUBSTRING(contact_name,1,3)) FROM customers;

---

3. Display average salary rounded to 2 decimal places. التقريب للعدد الذي تحدده هنا العدد ٢

SELECT **ROUND**(**AVG**(salary),2) FROM employees;

---

4. Display max and min salary. عرض أكبر قيمة وأقل قيمة

SELECT **MAX**(salary), MIN(salary) FROM employees;

---

5. Display year from order_date. استخراج السنة

SELECT **EXTRACT**(YEAR FROM order_date) FROM orders;

Joins & Relationships

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1. Display student name with class name.

```
SELECT s.name, c.name
FROM students s
JOIN classes c ON s.class_id = c.id;
```

---

2. Display student names with subjects.

```
SELECT s.name, sub.title
FROM students s
JOIN student_subjects ss ON s.id = ss.students_id
JOIN subjects sub ON ss.subject_id = sub.id;
```

---

3. Relationship between students and classes:
✅ One-to-Many

---

4. Relationship between students and subjects:
✅ Many-to-Many

Data Warehouse Architecture

1- Question: A global company has multiple systems (ERP, CRM, sales systems) and wants to unify data in one place for analysis. What architecture is most suitable for building a Data Warehouse? Discuss advantages and disadvantages.

Answer:

- **Solution:** Enterprise Data Warehouse (EDW – centralized architecture)
- **Reason:** The company needs to integrate all data from different sources and analyze it centrally.
- **Advantages:**
    1. Unified company-wide view.
    2. Easy cross-departmental analysis.
    3. Improved data quality and consistency.
- **Disadvantages:**
    1. High infrastructure cost.
    2. Long implementation time.
    3. Complexity in scaling if not properly planned.

2- Which approach is better: ETL or ELT? Explain advantages and disadvantages.

Answer:

- **Solution:** ELT (Extract, Load, Transform)
- **Reason:** For large datasets, it is better to load data first into the warehouse and transform it internally rather than transform before loading.
- **Advantages:**
    1. Faster loading process.
    2. Utilizes the high processing power of the data warehouse.
    3. Flexibility to modify transformations later.
- **Disadvantages:**
    1. Requires a powerful data warehouse.
    2. Internal transformations may affect performance.

3- Data Warehouse vs Data Mart

Question: A company wants to start analyzing only the sales department data before generalizing the system.

Answer:

- **Solution:** Data Mart
- **Reason:** Allows quick, low-cost analysis for one department before scaling.
- **Advantages:**
    1. Fast implementation.
    2. Lower cost.
    3. Suitable for testing and evaluation.
- **Disadvantages:**
    1. May cause data duplication during expansion.
    2. Does not provide a unified company-wide view initially.

4- Star Schema vs Snowflake Schema

Question: A company wants to design an analytical database that is easy and fast for querying.

Answer:

- **Solution:** Star Schema
- **Reason:** Simpler design and faster queries compared to Snowflake.
- **Advantages:**
    1. Easy to understand and design.
    2. Faster query performance for analytics.
- **Disadvantages:**
    1. Data redundancy.
    2. Can be harder to scale with very large datasets.

5- OLAP vs OLTP

Question: An e-commerce company wants a system to analyze customer behavior, not daily transactions.

Answer:

- **Solution:** OLAP (Online Analytical Processing)
- **Reason:** OLAP is designed for analysis and decision-making, whereas OLTP handles daily transactions.
- **Advantages:**
    1. Supports complex analytical queries.
    2. Enables multidimensional reporting.
- **Disadvantages:**
    1. Not suitable for daily transactional processing.
    2. Requires a well-structured data warehouse.

6- Incremental Load vs Full Load

Question: A company updates its data daily and wants to minimize loading time.

Answer:

- **Solution:** Incremental Load
- **Reason:** Loading only new or changed data reduces time and system load.
- **Advantages:**
    1. Faster than full load.
    2. Lower storage and processing cost.
- **Disadvantages:**
    1. Change management is more complex.
    2. Partial load failures may cause data issues.

7- Cloud Data Warehouse vs On-Premise

Question: A startup wants a fast, scalable solution without heavy infrastructure costs.

Answer:

- **Solution:** Cloud Data Warehouse
- **Reason:** Quick setup, no need for heavy infrastructure, and easily scalable.
- **Advantages:**
    1. Rapid deployment.
    2. Low initial cost.

3. Easily scalable.

- **Disadvantages:**
  1. Reliance on internet connectivity.
  2. Security concerns if not properly managed.
  3. Ongoing usage costs.

8-Real-time vs Batch Processing

Question: A company wants to analyze sales data instantly throughout the day.

Answer:

- **Solution:** Real-time Processing
- **Reason:** They need continuous updates for immediate decision-making.
- **Advantages:**
  1. Continuous data updates.
  2. Supports immediate decision-making.
- **Disadvantages:**
  1. Higher cost.
  2. More complex implementation and maintenance.

9-Centralized vs Decentralized Warehouse

Question: A company has multiple branches worldwide.

Answer:

- **Solution:** Centralized Data Warehouse
- **Reason:** Provides a unified view for global analysis.
- **Advantages:**
  1. Comprehensive view of the company.
  2. Easier control and data quality management.
- **Disadvantages:**
  1. Potential latency for remote branches.
  2. Higher data transfer costs.

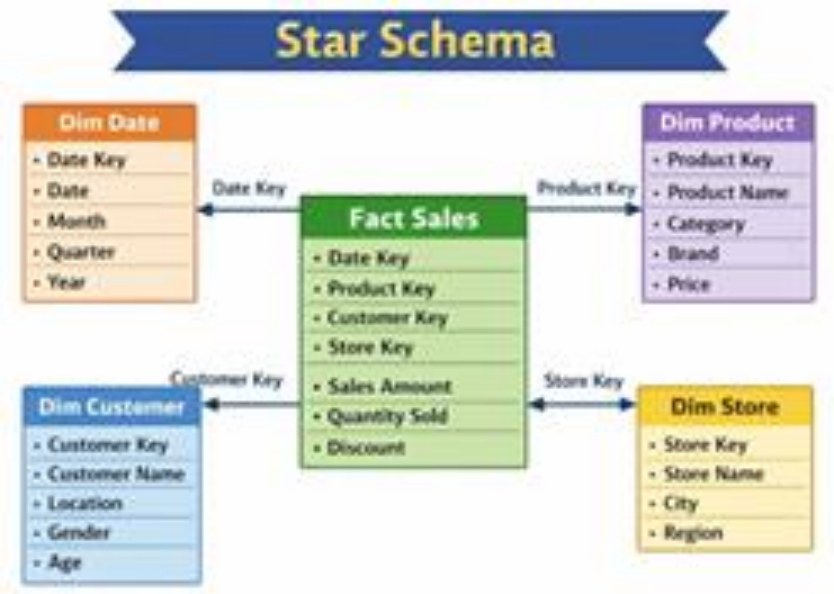
10-Data Lake vs Data Warehouse

Question: A company collects unstructured data (images, texts, logs).

Answer:

- **Solution:** Data Lake
- **Reason:** Can store all types of data (structured & unstructured) without transformation.
- **Advantages:**
  1. High flexibility in data storage.
  2. Supports analytics and machine learning later.
- **Disadvantages:**
  1. Data quality may be low.
  2. Searching and analysis require additional tools.
  3. Can become chaotic if not properly managed.

11-Star Schema



First: Multiple Choice Questions (MCQ)

1. What is the central table in a Star Schema?
A) Dimension Table
B) Fact Table ✅
C) Lookup Table
D) Bridge Table

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2. What does the Fact Sales table usually contain?
A) Descriptive data
B) Keys and measures (numeric values) ✅
C) Customer names
D) Dates only

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3. Which of the following is a measure?
A) Product Name
B) Customer Name
C) Sales Amount ✅
D) Store Location

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4. The relationship between the Fact Table and Dimension Tables is:
A) Many-to-Many
B) One-to-One
C) One-to-Many ✅
D) No relationship

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5. The main purpose of using a Star Schema is:
A) Data storage only
B) Improve query and analytical performance ✅
C) Reduce the number of tables
D) Delete data

Second: True or False

1. The Fact Table contains foreign keys.
✅ True

2. Dimension Tables contain only numeric data.
❌ False

3. Star Schema is used in Data Warehousing.
✅ True

4. Each Dimension Table is directly connected to the Fact Table.
✅ True

5. JOIN operations cannot be used in a Star Schema.
❌ False